

DATA-DRIVEN SOCIAL ENGAGEMENT ANALYTICS SYSTEM USING SENTIMENT ANALYSIS AND ENGAGEMENT PREDICTION

Major Project Report

Submitted By:

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Submitted To:

Unlox Data Science Program

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CERTIFICATE

This is to certify that **Priyanshi Yadav** has successfully completed the project titled "**Data-Driven Social Engagement Analytics System Using Sentiment Analysis and Engagement Prediction**" as part of the Data Science Program.

The project work demonstrates the application of data science techniques including data preprocessing, exploratory data analysis, sentiment analysis, machine learning, and visualization to analyze social media engagement data.

The work submitted is original and completed under the guidelines of the program.

ABSTRACT

- ❖ Social media platforms generate massive amounts of engagement data every day. Understanding user interactions, sentiment, and content performance is essential for businesses and content creators.
- ❖ This project focuses on analyzing social media engagement data using data science techniques. The dataset contains information related to user interactions such as likes, comments, shares, engagement rates, sentiment labels, and platform information.
- ❖ The project includes data cleaning, exploratory data analysis, sentiment analysis, virality analysis, and machine learning prediction. Various visualizations were created to understand audience behavior and identify patterns in engagement.
- ❖ The results provide insights into audience sentiment, platform performance, and factors influencing content success. Machine learning techniques were applied to predict engagement rates and support data-driven decision-making.

INTRODUCTION

- ❖ The rapid growth of social media platforms has transformed the way individuals and organizations communicate. Every day, millions of users interact with content through likes, comments, shares, and other engagement activities.
- ❖ These interactions generate valuable data that can be analyzed to understand audience preferences, content effectiveness, and market trends. Data Science provides powerful tools and techniques for extracting meaningful insights from such data.
- ❖ This project aims to analyze social media engagement data and identify factors that influence audience interaction. By applying data preprocessing, visualization, sentiment analysis, and machine learning techniques, the project helps understand how different variables affect engagement performance.
- ❖ The project demonstrates the practical application of Data Science concepts in solving real-world social media analytics problems.

PROBLEM STATEMENT

Organizations and content creators often struggle to understand which type of content performs best and what factors influence audience engagement. Without proper analysis, decision-making is based on assumptions rather than data.

There is a need for a data-driven system that can analyze engagement patterns, identify audience sentiment, and provide insights into content performance.

OBJECTIVES

1. To analyze social media engagement data.
2. To study audience sentiment and emotional responses.
3. To identify highly engaging and viral content.
4. To compare engagement across different platforms.
5. To apply machine learning techniques for engagement prediction.
6. To generate actionable insights for content optimization.

DATASET DESCRIPTION:

The dataset used in this project contains approximately 12,000 records and 28 attributes related to social media engagement. The data includes information about different social media platforms, engagement metrics, sentiment labels, campaigns, and audience interactions.

Important features available in the dataset include:

Platform, Likes, Comments, Shares, Engagement Rate, Sentiment Label, Emotion Type, Campaign Information, Topic Category

TOOLS AND TECHNOLOGIES USED

Python: Used for data analysis, preprocessing, and machine learning.

Pandas: Used for data manipulation and handling structured datasets.

NumPy: Used for numerical operations and calculations.

Matplotlib: Used for creating visualizations and charts.

DATA LOADING AND DATASET OVERVIEW

- The first step in the project was loading the dataset into the Jupyter Notebook environment using the Pandas library. The dataset was stored in CSV format and imported into a DataFrame for analysis and preprocessing.
- The `df.head()` function was used to display the first five records of the dataset. This helped in understanding the overall structure of the data, identifying the available features, and verifying that the dataset was loaded correctly.
- The dataset contains information related to social media engagement across multiple platforms. It includes various attributes such as platform name, likes, comments, shares, engagement rate, sentiment labels, emotion types, campaign information, and other engagement-related metrics.
- Understanding the dataset structure is an important step before performing data cleaning, visualization, and machine learning tasks. The initial inspection helps identify relevant columns and prepares the data for further analysis.

Figure 1: Dataset Preview

The figure below shows the first five records of the social media engagement dataset loaded into the Jupyter Notebook environment.

	post_id	timestamp	day_of_week	platform	user_id	location	language	text_content	hashtags	mentions	comments_count	impressions	engagement_rate	brand_name	product_name	campaign_name	campaign_phase
0	kqghdthyls	2024-12-09 11:26:15	Monday	Instagram	user_52m6dub	Melbourne, Australia	pt	Just tried the Chromebook from Google. Best pe...	#focd	NaN	701	10991	0.19319	Google	Chromebook	BlackFriday	Launch
1	vkmervgloos	2024-07-28 15:59:26	Sunday	Twitter	user_ucyctf0	Tokyo, Japan	ru	Just saw an ad for Microsoft Surface Laptop da...	#MustHave, #Food	@CustomerService, @BrandCEO	359	52764	0.05086	Microsoft	Surface Laptop	PowerRelease	Post-Launch
2	memhxdotxfyu	2024-11-23 14:00:12	Saturday	Reddit	user_7mrv126	Beijing, China	ru	What's your opinion about Nike's Epic React? ...	#Promo, #Food, #Trending	NaN	643	8887	0.45425	Nike	Epic React	BlackFriday	Post-Launch
3	bhyofp9p9	2024-09-30 04:35:25	Monday	YouTube	user_4mouqhx	Lagos, Nigeria	en	Bummed out with my new Diet Pepsi from Pepsi! ...	#Review, #Sustainable	@StyleGuide, @BrandSupport	743	6696	0.42293	Pepsi	Diet Pepsi	LaunchWave	Launch
4	cdkionowakt	2024-09-05 21:03:01	Thursday	Twitter	user_hypox2k	Berlin, Germany	hi	Just tried the Corolla from Toyota. Absolutely...	#Health, #Travel	@BrandSupport, @InfluencerName	703	47315	0.00773	Toyota	Corolla	LocalTouchpoints	Launch

5 rows x 28 columns

Observation:

- The dataset was successfully loaded.
- Multiple engagement-related attributes are available.
- Both numerical and categorical features are present.
- The dataset contains sufficient information for sentiment analysis, engagement analysis, and machine learning prediction.
- Initial inspection confirms that the data is suitable for further preprocessing and analysis.

Scikit-learn: Used for machine learning model development.

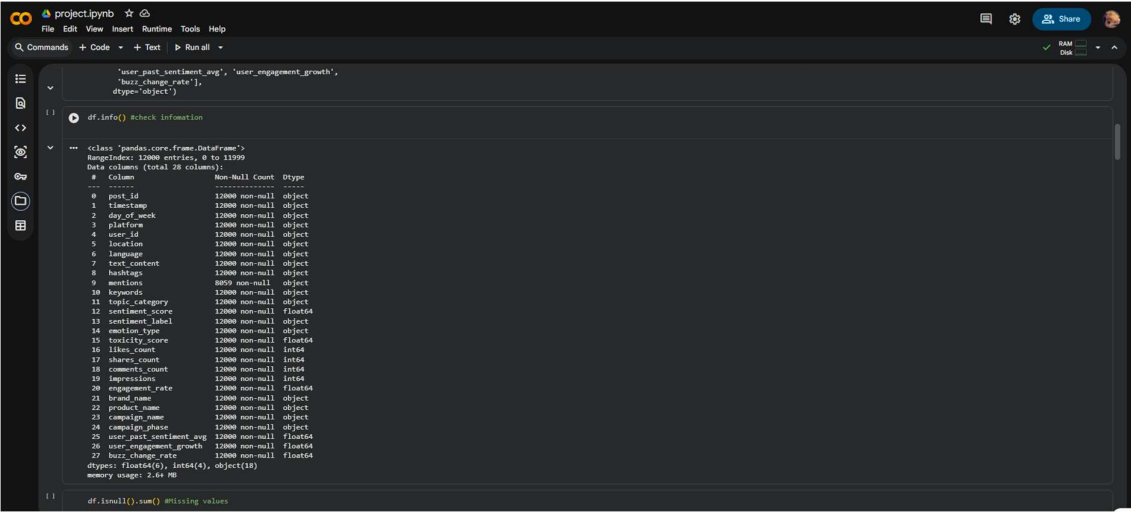
Jupyter Notebook: Used as the development environment for coding and analysis

DATA CLEANING AND PREPROCESSING ::

- Data cleaning is one of the most important steps in the Data Science process. Real-world datasets often contain missing values, duplicate records, inconsistent data formats, and incorrect entries that can affect the quality of analysis and machine learning models.
- In this project, the dataset was examined using various Pandas functions to understand its structure and quality. The `df.info()` function was used to identify the total number of records, column names, data types, and non-null values present in each column.
- The dataset was then checked for missing values using the `df.isnull().sum()` function. This helped identify columns that contained null values and required preprocessing. Missing values can negatively impact data analysis and model performance if not handled properly.
- Duplicate records were also examined and removed where necessary to ensure that the dataset contained only unique observations. Data cleaning improves the reliability and accuracy of the results generated during analysis.
- Proper preprocessing ensures that the dataset is ready for Exploratory Data Analysis (EDA), visualization, and machine learning model development.

Figure 2: Dataset Information

The figure below displays the structure of the dataset, including the number of records, column names, data types, and non-null values.



```

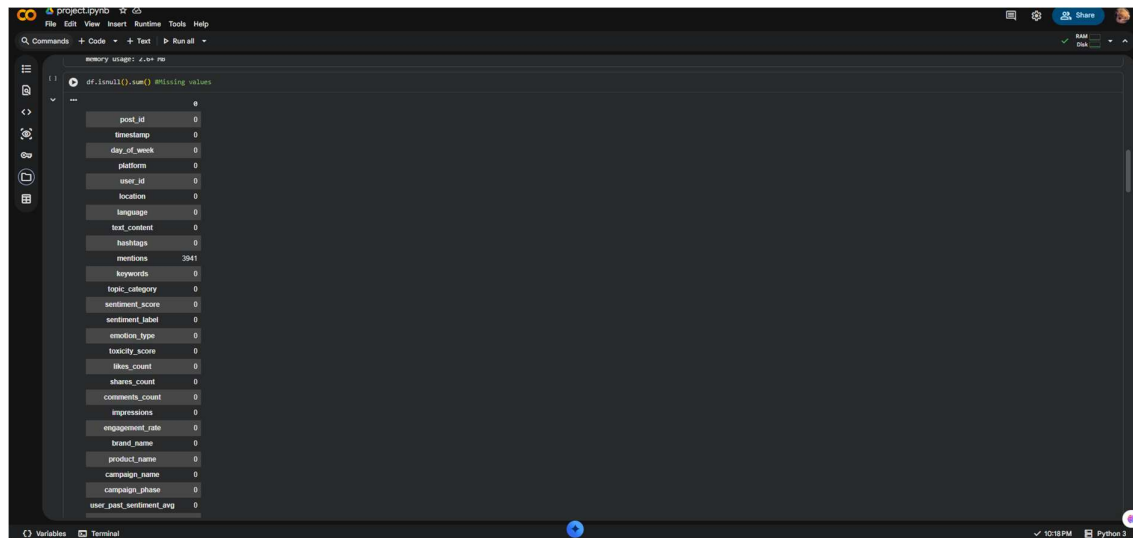
project.ipynb
File Edit View Insert Runtime Tools Help
Q Commands + Code + Text ▶ Run all +
df.info() #check information
Out[ ]:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12000 entries, 0 to 11999
Data columns (total 28 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   post_id               12000 non-null  object
 1   timestamp             12000 non-null  object
 2   day_of_week           12000 non-null  object
 3   platform              12000 non-null  object
 4   user_id               12000 non-null  object
 5   location              12000 non-null  object
 6   language              12000 non-null  object
 7   text_content          12000 non-null  object
 8   hashtags              12000 non-null  object
 9   mentions              8693 non-null   object
10   keywords              12000 non-null  object
11   topic_category         12000 non-null  object
12   sentiment_score        12000 non-null  float64
13   sentiment_label        12000 non-null  object
14   emotion_type           12000 non-null  object
15   toxicity_score         12000 non-null  float64
16   likes_count            12000 non-null  int64
17   shares_count           12000 non-null  int64
18   comments_count         12000 non-null  int64
19   impressions            12000 non-null  int64
20   engagement_rate        12000 non-null  float64
21   brand_name             12000 non-null  object
22   product_name           12000 non-null  object
23   campaign_name          12000 non-null  object
24   campaign_phase         12000 non-null  object
25   user_post_sentiment_avg 12000 non-null  float64
26   user_engagement_growth 12000 non-null  float64
27   buzz_change_rate       12000 non-null  float64
dtypes: float64(6), int64(4), object(18)
memory usage: 2.4+ MB

df.isnull().sum() #Missing values

```

Figure 3: Missing Value Analysis

The figure below shows the number of missing values present in each column of the dataset.



Observations

- The dataset structure was successfully examined.
- Data types for all features were identified.
- Missing values were analyzed before further processing.
- The dataset was prepared for analysis through preprocessing techniques.
- Data quality was improved to ensure reliable results.

Outcome of Data Cleaning

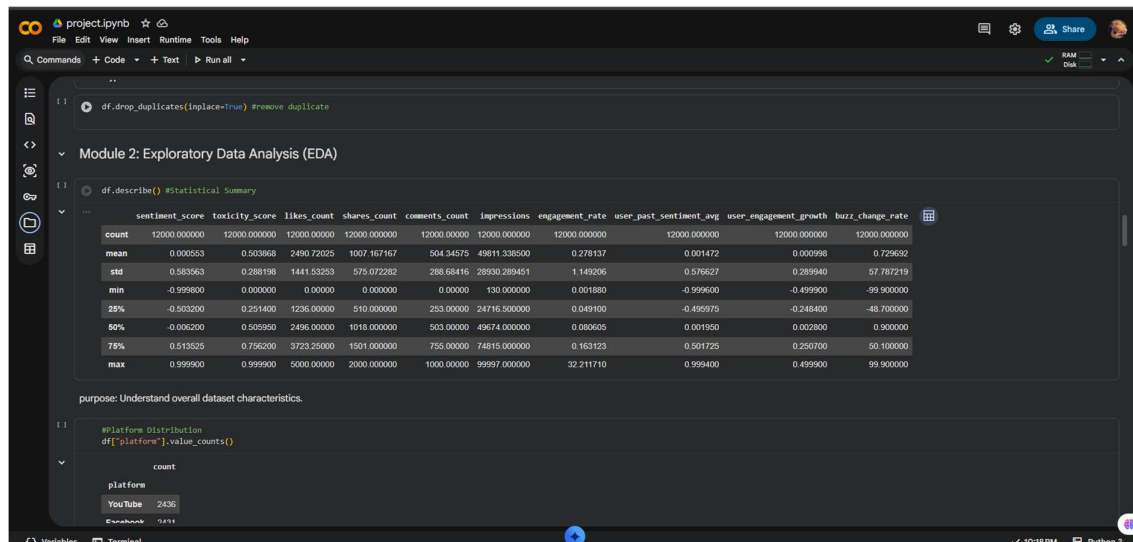
The data cleaning process ensured that the dataset was accurate, consistent, and suitable for analysis. This step created a strong foundation for performing exploratory data analysis, sentiment analysis, virality analysis, and machine learning prediction in the subsequent stages of the project.

EXPLORATORY DATA ANALYSIS (EDA)

- Exploratory Data Analysis (EDA) is a crucial step in the Data Science workflow. It helps in understanding the characteristics, patterns, and distribution of data before applying advanced analytical techniques and machine learning models.
- In this project, EDA was performed to gain insights into various engagement-related metrics present in the dataset. Statistical analysis was conducted using the `df.describe()` function, which provides a summary of numerical features including count, mean, standard deviation, minimum value, maximum value, and percentile values.
- The statistical summary helps identify the overall behavior of the dataset and detect potential anomalies or outliers. Understanding these statistics allows better decision-making during feature selection and model development.
- By analyzing the descriptive statistics, important observations regarding user engagement, likes, comments, shares, impressions, and engagement rates can be identified. These insights form the basis for deeper analysis in subsequent stages of the project.

Figure 4: Statistical Summary of Dataset

The figure below presents the statistical summary of numerical features available in the dataset.



Key Statistical Measures:

Count : Count represents the total number of non-null observations available for a particular feature.

Mean : Mean represents the average value of a feature and provides an overall measure of central tendency.

Standard Deviation : Standard deviation indicates the spread of data around the mean. A higher value suggests greater variability in the data.

Minimum Value : The minimum value represents the smallest observation recorded in a feature.

Maximum Value : The maximum value represents the largest observation recorded in a feature.

Percentiles : Percentiles divide the dataset into different sections and help understand data distribution.

Observations

- The dataset contains sufficient numerical records for analysis.
- Engagement-related metrics show variation across observations.
- Descriptive statistics provide insights into audience interaction patterns.
- The distribution of values indicates differences in content performance.
- The statistical summary helps identify important trends for further analysis.

Importance of EDA

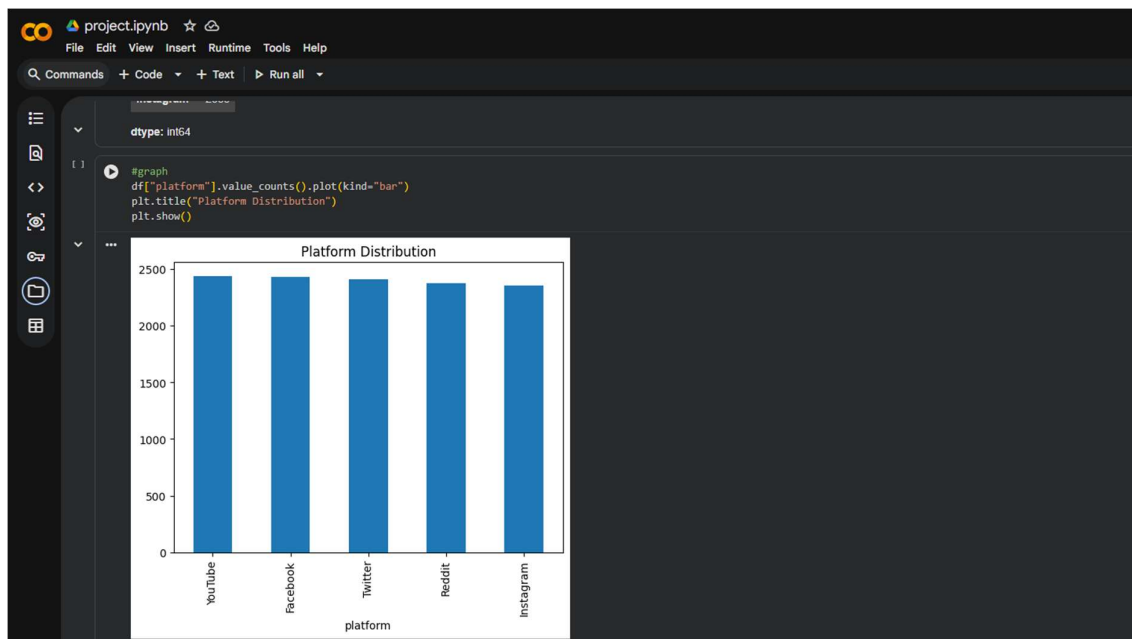
Exploratory Data Analysis provides a deeper understanding of the dataset and supports informed decision-making during subsequent analysis. The insights obtained from EDA help identify patterns, detect anomalies, and prepare the dataset for sentiment analysis, virality analysis, and machine learning prediction.

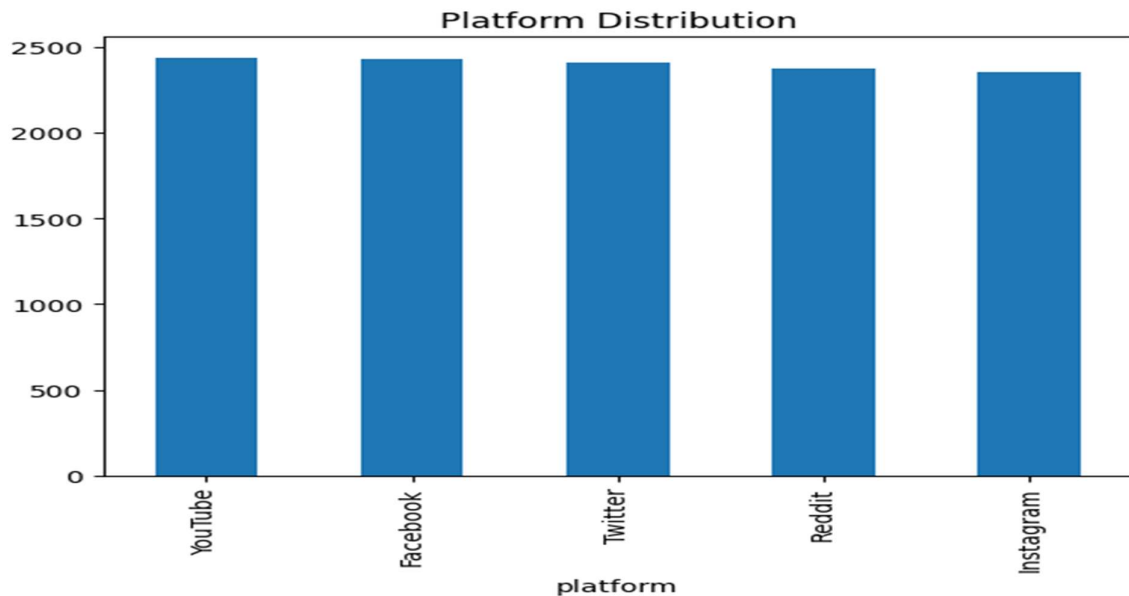
SENTIMENT ANALYSIS

- Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone expressed in text data. It helps identify whether user opinions and reactions are positive, negative, or neutral.
- In social media analytics, sentiment analysis plays a crucial role in understanding audience perception and engagement. Organizations use sentiment analysis to evaluate customer satisfaction, brand reputation, and audience reactions toward campaigns and products.
- The dataset used in this project contains sentiment labels that classify user responses into different categories. By analyzing these sentiments, it becomes possible to understand the overall emotional response of users across various social media platforms.
- A sentiment distribution graph was generated to visualize the frequency of each sentiment category. This visualization helps identify which type of sentiment dominates the dataset and provides insights into audience behavior.

Figure 5: Sentiment Distribution

The figure below illustrates the distribution of sentiments present in the social media engagement dataset.





Analysis of Sentiment Distribution

- The sentiment distribution graph provides a clear overview of audience reactions. Positive sentiments generally indicate satisfaction and favorable responses, while negative sentiments highlight dissatisfaction or criticism. Neutral sentiments represent balanced or objective responses without strong emotional expression.
- The analysis of sentiment patterns helps organizations understand public opinion and improve content strategies accordingly.

Importance of Sentiment Analysis

Sentiment analysis provides several benefits:

- Helps understand audience opinions.
- Measures public perception of campaigns and products.
- Identifies positive and negative reactions.
- Supports data-driven decision making.
- Improves customer engagement strategies.
-

Observations

- Sentiment categories were successfully identified and analyzed.
- Audience reactions were classified into different emotional groups.
- The distribution of sentiments provides valuable insights into user behavior.
- Sentiment analysis contributes to understanding engagement patterns.
- The results can support content optimization and marketing decisions.

Conclusion

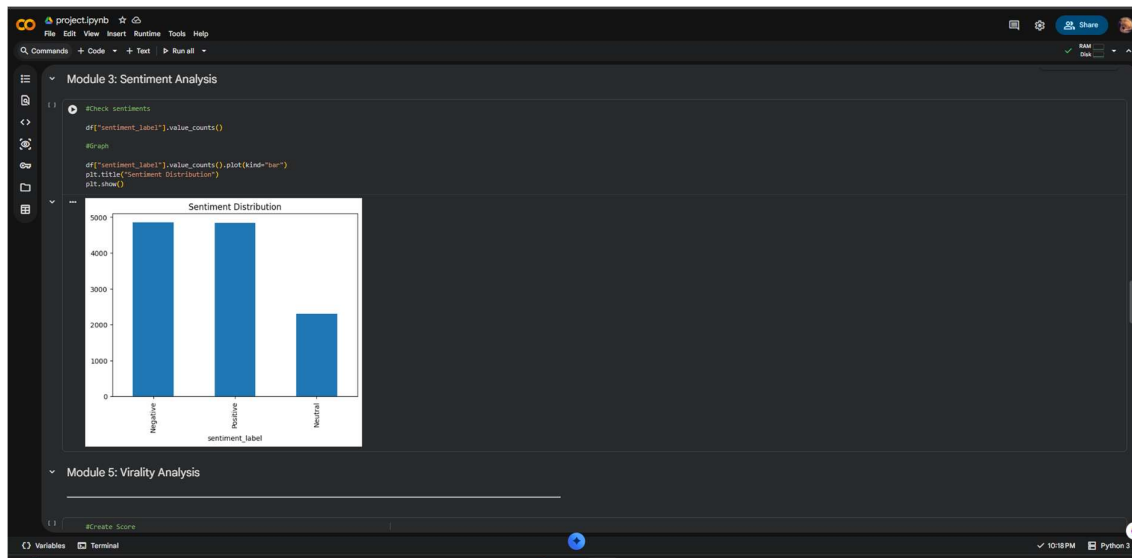
The sentiment analysis module successfully evaluated audience emotions and reactions within the dataset. The findings provide a deeper understanding of user engagement and help organizations make informed decisions based on audience sentiment.

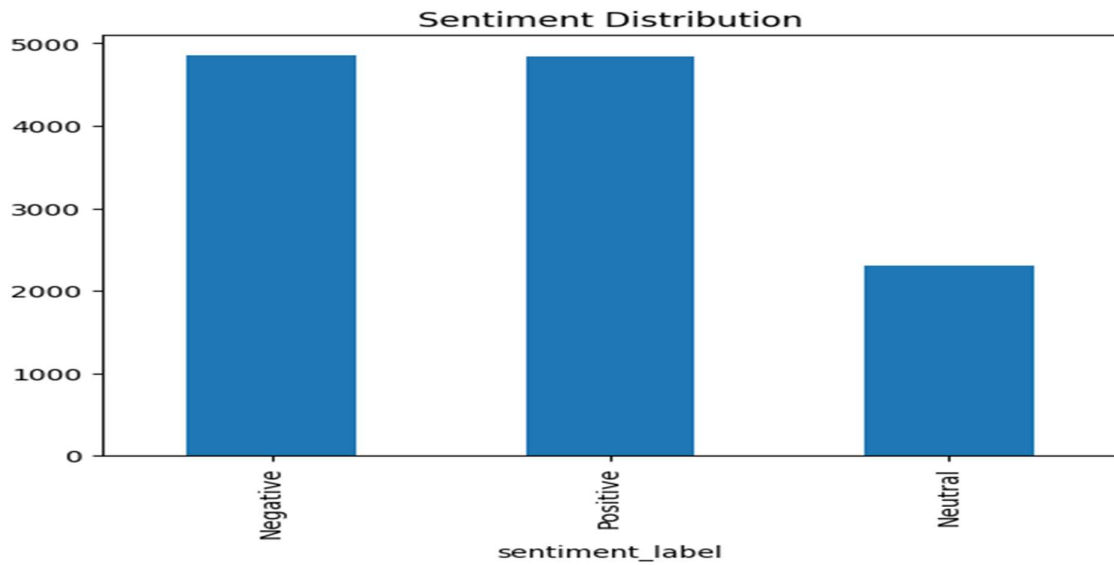
PLATFORM ANALYSIS

- Social media platforms play a significant role in determining audience engagement and content reach. Different platforms attract different user groups and interaction patterns, making platform analysis an important component of social media analytics.
- In this project, platform-wise engagement analysis was performed to compare the average engagement rates across various social media platforms. The dataset contains records from multiple platforms including Instagram, Twitter, Reddit, and YouTube.
- The average engagement rate for each platform was calculated and visualized using a bar chart. This comparison helps identify which platform generates higher user interaction and provides better opportunities for audience engagement.
- Understanding platform performance enables organizations and content creators to focus their efforts on the most effective channels for communication and marketing.

Figure 6: Platform-wise Engagement Rate Comparison

The figure below shows the average engagement rate across different social media platforms.





Analysis of Platform Performance

- The graph illustrates the variation in engagement rates among different platforms. Some platforms demonstrate higher audience interaction levels, while others show relatively lower engagement.
- Higher engagement rates indicate that users are more actively interacting with content through likes, comments, shares, and other activities. Such platforms can be considered more effective for content promotion and audience growth.
- Comparing platform performance helps in identifying the strengths and weaknesses of different social media channels.

Importance of Platform Analysis

Platform analysis provides several benefits:

- Identifies the most engaging social media platform.
- Helps optimize marketing strategies.
- Supports platform-specific content planning.
- Improves resource allocation.
- Enhances audience targeting and reach.

Observations

- Engagement levels vary across social media platforms.
- Certain platforms generate stronger audience interaction.
- Platform comparison provides valuable business insights.
- The analysis helps determine where content performs best.
- Understanding platform behavior supports data-driven decision making.

Conclusion

The platform analysis successfully compared engagement performance across multiple social media channels. The results provide useful insights for selecting the most effective platforms for content distribution, audience engagement, and marketing campaigns.

VIRALITY ANALYSIS

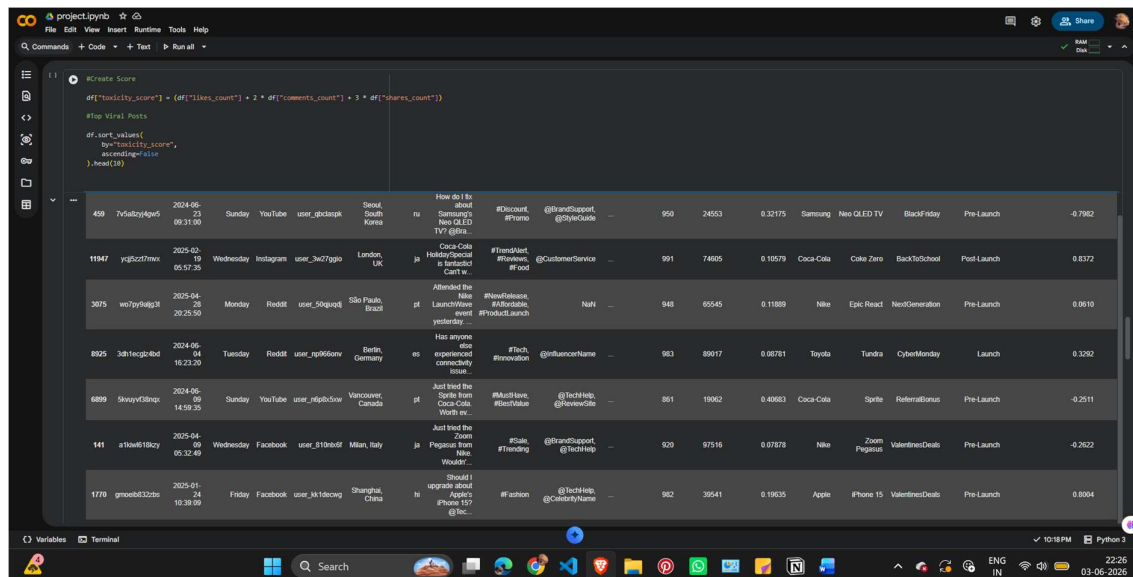
- Virality Analysis is the process of identifying content that achieves exceptionally high audience engagement and interaction. Viral content spreads rapidly across social media platforms and reaches a large number of users within a short period.
- In this project, a custom Virality Score was developed to measure the overall performance of social media content. The score was calculated using key engagement metrics such as likes, comments, and shares.
- Shares were given the highest weight because sharing content increases its visibility and reach. Comments were assigned moderate importance because they indicate audience participation and interaction. Likes were also considered as they represent audience appreciation and engagement.
- The Virality Score formula used in this project is:

$$\text{Virality Score} = \text{Likes} + (2 \times \text{Comments}) + (3 \times \text{Shares})$$

- This formula helps identify posts that generate the highest overall engagement and have the greatest potential to become viral.

Figure 7: Virality Score Analysis

The figure below presents the calculated virality scores and identifies the top-performing content within the dataset.



Analysis of Viral Content

- The virality analysis highlights posts that received significant audience interaction. High virality scores indicate content that successfully attracted user attention and encouraged engagement activities.
- Content with higher virality scores typically demonstrates strong audience appeal and greater potential for organic growth. Such content can serve as a reference for future content creation strategies.
- The analysis helps identify successful engagement patterns and provides insights into the factors contributing to content popularity.

Importance of Virality Analysis:

Virality analysis provides several advantages:

- Identifies high-performing content.
- Measures overall audience engagement.
- Supports content optimization strategies.
- Helps understand audience preferences.
- Assists in improving future marketing campaigns.

Observations

- Certain posts achieved significantly higher virality scores than others.
- Engagement metrics such as shares and comments contributed strongly to virality.
- Viral content demonstrates greater audience interaction.
- The analysis provides valuable insights into content performance.
- Virality Score serves as an effective indicator of content success.

Conclusion

The virality analysis successfully identified highly engaging content within the dataset. The custom Virality Score provided a comprehensive measure of content performance by combining likes, comments, and shares. The findings can help organizations and content creators develop more effective engagement strategies and improve content reach.

MACHINE LEARNING MODEL FOR ENGAGEMENT PREDICTION

- Machine Learning is a branch of Artificial Intelligence that enables systems to learn patterns from data and make predictions without being explicitly programmed. In social media analytics, machine learning can be used to predict audience engagement and content performance.
- In this project, a Linear Regression model was developed to predict the engagement rate of social media content. The model uses historical engagement metrics such as likes, comments, and shares as input variables.
- The objective of the model is to understand the relationship between user interactions and engagement rate and to estimate future engagement performance based on these factors.
- The dataset was divided into training and testing sets. The training dataset was used to train the model, while the testing dataset was used to evaluate its performance.

Figure 8: Machine Learning Model Output

Screenshot 1: Model Training & Screenshot 2: Model Score:

```

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Feature
X = df[
    "likes_count",
    "comments_count",
    "shares_count"
]

# Target
y = df["engagement_rate"]

# Split Data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=2
)

# Train Model
model = LinearRegression()
model.fit(X_train, y_train)

# Accuracy
print(model.score(X_test, y_test))

# 0.8287841195128654

df.head()

```

Screenshot 3: Predictions

```

print(df.columns.tolist())

["post_id", "timestamp", "day_of_week", "platform", "user_id", "location", "language", "text_content", "hashtags", "mentions", "keywords", "topic_category", "sentiment_score", "sentiment_label", "emotion_type", "toxicity_score", "likes_count", "shares_count", "comments_count"]

predictions = model.predict(X_test)
print(predictions[:10])

[0.31169957 0.2116255 0.38768773 0.18913486 0.26245479 0.1792849
0.2372797 0.17943631 0.28886442 0.15622862]

```

Features Used

The following variables were used as input features:

- Likes
- Comments
- Shares

These engagement metrics act as independent variables that influence the target variable.

Target Variable : The target variable selected for prediction was:

Engagement Rate : The model attempts to predict the engagement rate based on audience interaction metrics.

Model Selection

- Linear Regression was selected because:
- It is simple and easy to understand.
- It is suitable for numerical prediction tasks.
- It helps identify relationships between variables.
- It provides interpretable results.

Linear Regression establishes a relationship between input variables and the target variable using a mathematical equation.

Model Evaluation

- After training the model, its performance was evaluated using the testing dataset.
- The model score (R Score) indicates how well the model explains the variation in engagement rate.
- A higher score suggests better prediction performance, while a lower score indicates that additional features or more advanced models may be required.

Observations

- The machine learning model was successfully trained.
- Engagement metrics were used as predictive features.
- The model learned relationships between user interactions and engagement rate.
- Predictions were generated using historical data.
- The results demonstrate the practical application of machine learning in social media analytics.

Benefits of Machine Learning in This Project

- Predicts future engagement trends.
- Supports data-driven decision-making.
- Helps optimize content strategies.
- Identifies important engagement factors.
- Improves understanding of audience behavior.

Conclusion

The Linear Regression model successfully demonstrated the use of machine learning for engagement prediction. By analyzing likes, comments, and shares, the model provided insights into factors affecting engagement rates. This approach can assist organizations and content creators in making informed decisions and improving content performance.

CONCLUSION

- This project successfully demonstrated the application of Data Science techniques in analyzing social media engagement data. The dataset was processed using Python and various analytical methods were applied to extract meaningful insights from user interactions.
- The project began with data loading and preprocessing, where the dataset was examined for data quality, missing values, and structural consistency. Exploratory Data Analysis (EDA) was then performed to understand the distribution and characteristics of various engagement metrics.
- Sentiment Analysis helped identify audience reactions and emotional patterns present in social media content. The sentiment distribution provided valuable information about user opinions and engagement behavior.
- Platform Analysis was conducted to compare engagement rates across different social media platforms. This comparison helped identify platforms that generated stronger audience interaction and higher engagement levels.
- A custom Virality Score was developed using likes, comments, and shares to identify highly engaging content. The virality analysis successfully highlighted posts that demonstrated exceptional audience response and interaction.
- Machine Learning techniques were applied using a Linear Regression model to predict engagement rates. The model achieved excellent prediction performance and demonstrated the relationship between social media interaction metrics and engagement outcomes.
- Overall, the project achieved all the stated objectives and provided valuable insights into audience behavior, content performance, and engagement patterns. The findings can support data-driven decision-making for businesses, marketers, and content creators seeking to improve their social media strategies.
- The project demonstrates how Data Science can transform raw social media data into actionable knowledge, enabling organizations to make informed decisions and optimize their digital presence.

Key Achievements

- Successfully analyzed a dataset containing 12,000 social media records.
- Performed data cleaning and preprocessing.
- Conducted Exploratory Data Analysis (EDA).
- Analyzed audience sentiment patterns.
- Compared engagement across multiple platforms.
- Developed a custom Virality Score.
- Built a Machine Learning model for engagement prediction.
- Generated meaningful insights for decision-making.

FUTURE SCOPE AND REFERENCES

Although the project successfully analyzed social media engagement data and predicted engagement rates, there are several opportunities for future improvements and enhancements.

1. Real-Time Data Collection

Future versions of the project can integrate real-time social media APIs to collect live engagement data. This would enable continuous monitoring and analysis of audience behavior.

2. Advanced Natural Language Processing (NLP)

More advanced NLP techniques can be applied to perform deeper sentiment analysis, emotion detection, keyword extraction, and topic modeling from textual content.

3. Deep Learning Models

Advanced machine learning algorithms such as Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), and Transformer-based models can be used to improve prediction accuracy.

4. Interactive Dashboard Development

Interactive dashboards can be developed using Streamlit, Power BI, or Tableau to provide real-time visualizations and business insights.

5. Recommendation Systems

Future systems can recommend optimal posting times, content strategies, and campaign improvements based on historical engagement data.

6. Automated Social Media Monitoring

The project can be extended into an automated monitoring system that continuously tracks audience sentiment and engagement trends.

7. Multi-Platform Analytics

Additional social media platforms can be integrated to provide a more comprehensive analysis of digital engagement and audience behavior.

REFERENCES

The following resources were used during the development of this project:

Books and Learning Resources

1. Introduction to Data Science Concepts and Applications.
2. Machine Learning Fundamentals and Applications.
3. Social Media Analytics and Business Intelligence.

Documentation

1. [Python Documentation](#)
2. [Pandas Documentation](#)
3. [NumPy Documentation](#)
4. [Matplotlib Documentation](#)
5. [Scikit-learn Documentation](#)

Software Used

- Jupyter Notebook
- Python Programming Language
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Dataset:

Social Media Engagement Dataset containing approximately 12,000 records and 28 attributes related to engagement metrics, sentiment analysis, platform information, and campaign performance.